

Detection and Analysis of Flipper Zero Deauthentication Signals using HackRF One Software-defined radio

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Abstract— Deauthentication attacks remain a significant threat to wireless network security. This study investigates the characteristics of deauthentication signals generated by a Flipper Zero device as captured by a HackRF One Software-defined radio (SDR). By analyzing signal parameters such as power levels, frequency bands, and modulation techniques, distinct features differentiating deauthentication attacks from legitimate network traffic are identified. Results indicate that deauthentication attacks introduce notable power level spikes, increasing from approximately -80 dBm to -30 dBm at a distance of one meter. The deauthentication signal typically occupies a 6 MHz bandwidth centered around 2.422 GHz. Signal strength decreases to -35 dBm and -40 dBm at distances of five and ten meters, respectively. These findings complement existing packet-based detection methods by providing additional insights into deauthentication attack characteristics, enabling the development of more robust detection systems through the integration of waveform analysis and advanced signal processing techniques.

Keywords—flipper-zero, deauthentication, software-define radio, wireless, network, access point

I. INTRODUCTION

Wireless networks have become an integral part of modern infrastructure, supporting a vast array of critical applications. Wi-Fi technology has evolved over the years, enabling high-speed wireless communication and connectivity for a wide range of consumer devices such as smartphones, tablets, laptops, smart TVs, and IoT devices [1]. However, their reliance on open radio channels introduces inherent vulnerabilities to security breaches. Deauthentication attacks, where malicious actors employ deauthorization packets to disrupt legitimate user connections, pose a significant threat to network stability [2].

The Flipper Zero, a compact device equipped with a radio transceiver and various interfaces by Flipper Devices Inc, has emerged as a concern due to its potential for misuse in Wi-Fi network attacks. While offering legitimate functionalities like Radio-frequency identification (RFID) and Near-field communication (NFC) interaction [3], add-on boards and modified firmware can transform it into a tool for launching denial-of-service assaults. By flooding access points with deauthorization packets, the Flipper Zero disrupts active user connections, hindering network operations [4].

This paper presents an investigation into the technical aspects of the Flipper Zero, focusing on its deauthentication attack capabilities. This study leverage the HackRF On SDR of Great Scott Gadgets for detection and analysis of deauth signals generated by the Flipper Zero. The captured information from these signals serves as the foundation for developing a system or device capable of identifying and

mitigating deauthentication attacks launched using the Flipper Zero and similar portable hacking devices.

II. EXPERIMENTAL SET UP

A. Main Devices

Flipper Zero is a multi-purpose hacking tool designed for interacting with various protocols like RFID, NFC, and infrared [5] as shown on Figure 1. It can also be used for debugging and flashing firmware. It does not come equipped with integrated Wi-Fi functionality.

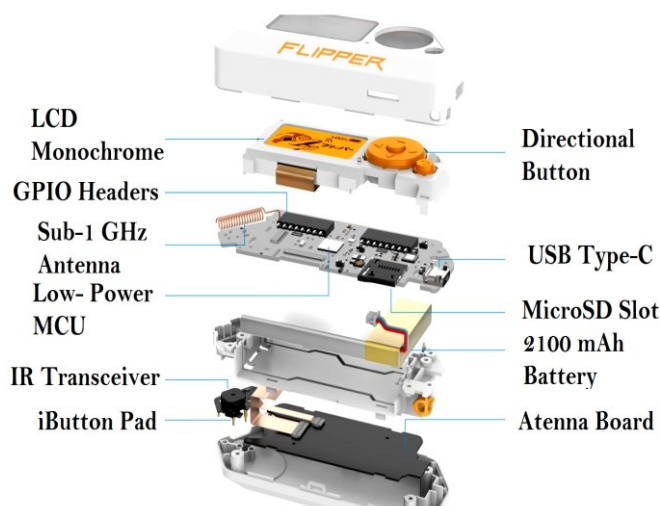


Fig. 1. Inside a Flipper Zero [5]

However, the Wi-Fi Dev Board, an add-on module utilizing the ESP32-S2 chip, significantly enhances the Flipper Zero by providing wireless capabilities. This expansion allows for features such as wireless firmware updates for the Flipper Zero itself and advanced in-circuit debugging via Wi-Fi [6]. The Developer Board is equipped with an ESP32-S2-WROVER module, which includes built-in Wi-Fi capabilities. It also offers GPIO pins for easy connectivity to various targets. Additionally, the Developer Board features a USB Type-C connector for data transfer and power supply. For user interaction, the Developer Board has tactile switches. Figure 2 illustrates the development board connected to the Flipper Zero.

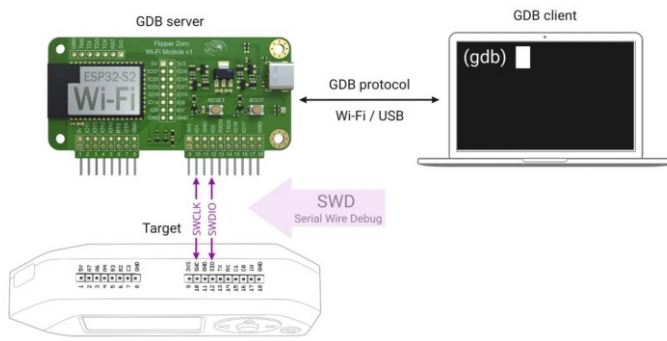


Fig. 2. Wi-Fi Developer Board connected to Flipper Zero [6]

The HackRF One with Portapack H2 is a versatile SDR platform designed for a wide range of wireless communication applications. The HackRF One covers a frequency range of 1 MHz to 6 GHz and supports half-duplex operation, enabling both the transmission and reception of signals. It can operate as a USB peripheral or programmed for stand-alone operation, making it highly adaptable for various uses such as spectrum analysis, signal identification, and radio communication experimentation. Figure 3 shows the actual image of the HackRF One with the Portapack H2 module attached, highlighting its compact design and user-friendly interface.

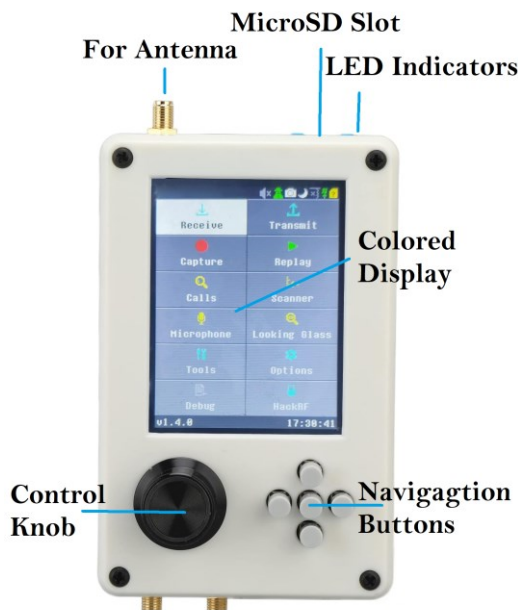


Fig. 3. HackRF One Portapack H2 [7]

Tests were conducted using a dual-band Wi-Fi antenna that operates across frequencies of 2.4 GHz (2400-2485 MHz) and 5GHz/5.8 GHz (5150-5850 MHz) with a 3 dBi gain. Featuring an RP-SMA male connector, these antennas are compatible with a wide range of devices, including wireless network routers, Wi-Fi USB adapters, desktop PC wireless network cards, IP security cameras, and various video surveillance systems.

Deauthentication attack is tested using a test network with the EG8145X6-10 as the main access point. The EG8145X6-10 is an optical network terminal (ONT) designed for fiber-to-the-home (FTTH) networks. It typically supports Gigabit Passive Optical Network (GPON) technology, providing high-speed internet, voice, and video services over a single fiber optic connection.

B. Software

The HackRF Spectrum Analyzer is a specialized software tool designed for utilizing the HackRF One as its waveform analyzer, particularly useful in the detection and analysis of signals such as those generated by the Flipper Zero tool. It features automatic restart of `hackrf_sweep` with any settings changes, easy retuning, peak/persistent display, frequency allocation bands, a high-resolution waterfall plot and a spur filter to remove artifacts from the spectrum.

Cubic SDR is also used as the main software tool. It is a versatile cross-platform SDR application designed for real-time signal processing and analysis. Utilizing libraries such as liquid-digital signal processing and Soapy SDR, Cubic SDR supports a wide range of hardware. CubicSDR provides a robust platform to capture, visualize, and decode signals, including the detection and analysis of deauthentication signals.

C. Firmware

Xtreme Firmware (XFW) for Flipper Zero enhances the device with additional features, improving its capabilities for various tasks, including Wi-Fi related attacks. When installed on Flipper Zero, this firmware enables users to execute and analyze Wi-Fi deauthentication signals with greater efficiency, making it an invaluable tool for the examination of wireless security threats and vulnerabilities.

The Marauder firmware for Wi-Fi development boards is a highly versatile and powerful suite that equips users with a range of both offensive and defensive tools tailored for network security tasks. Developed specifically for these platforms, it offers robust capabilities such as scanning for available Wi-Fi networks, monitoring network traffic, and launching targeted attacks like deauthentication, packet injection, and handshake capturing. This firmware is not only used for offensive security testing and network penetration but also for defensive purposes, allowing users to assess vulnerabilities and strengthen the security strength of Wi-Fi environments.

The HackRF One Porta Pack H2 Mayhem firmware is an enhanced version of the Havoc firmware, which itself was an extension of the original Porta Pack firmware. This firmware enables extensive Software-defined radio (SDR) functionality, such as signal analysis and transmission capabilities.

D. Test Set up

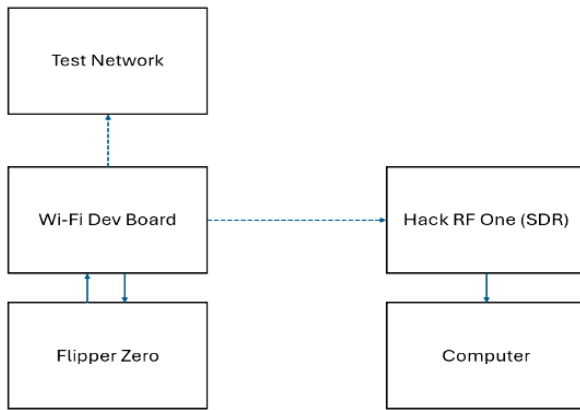


Fig. 4. Block Diagram of the Test Set up

Figure 4 illustrates the block diagram of the test setup, where the Wi-Fi Development Board is connected to Flipper Zero. In this setup, the HackRF One sweeps the 2.4 to 2.45 GHz frequency band to receive signals, which are then transmitted serially to a computer. The computer is equipped with HackRF Spectrum Analyzer software, enabling detailed analysis

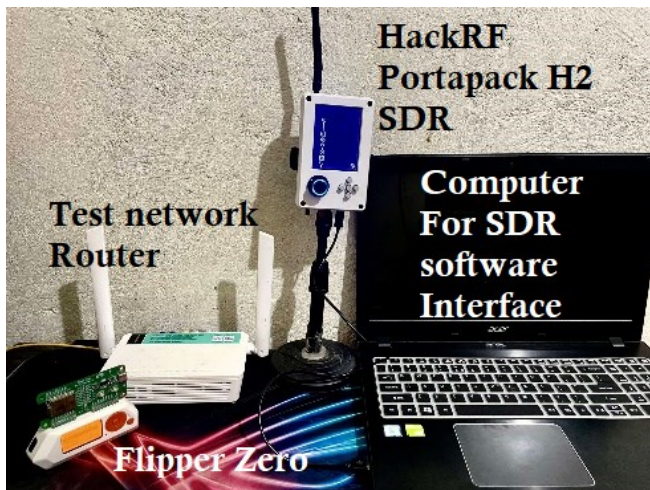


Fig. 5. Actual Image of the Test Set up

The Wi-Fi Development Board, connected to the Flipper Zero, facilitates the generation of Wi-Fi deauthentication signals, while the HackRF One captures these signals for analysis. The actual set up is shown in Figure 5.

PROCEDURE

Firmware was installed on the Flipper Zero, Wi-Fi Development Board, and HackRF One. The HackRF Spectrum Analyzer and Q-Flipper software were installed on a computer. A test network was set up using the EG8145X6-10, and surrounding networks were scanned to select the test network. A deauthentication attack was launched using the Flipper Zero. The deauthentication signal was immediately captured within a few seconds by the SDR. However, to ensure the capture of a stable signal image and account for any fluctuations, the attack was set to run for an average of 5 minutes during each trial.

HackRF One was connected to the computer serially. HackRF mode was selected, followed by the `hackrf_sweep` on the spectrum analyzer. 2.4 GHz as Start and 2.45 GHz as end frequencies were set, and the sweep was initiated. The Flipper Zero was placed within one meter of the HackRF One. Waveforms were captured over at least 25 trials, and measurements of lower, upper, and center frequency, as well as dB power levels, were tabulated. The procedure was repeated with the Flipper Zero placed five meters away from the SDR. The procedure was repeated again with the Flipper Zero placed ten meters away from the SDR.

III. RESULTS AND DISCUSSION

This section presents the results of an experimental analysis of deauthentication attacks, focusing on power level variations and signal characteristics. The discussion highlights the implications of these findings for the detection and mitigation of such attacks.

A. Power Level Spikes at Close Proximity

The experimental analysis reveals that deauthentication attacks introduce significant power level spikes. At a proximity of one meter, the power levels increase dramatically from approximately -80 dBm to -30 dBm. The deauthentication signal typically occupies a bandwidth of 6 MHz, centered around 2.422 GHz. Data were tabulated as shown on Table I.

TABLE I. DEAUTHENTICATION SIGNAL DATA CAPTURED WITHIN 1 METER

No	Deauthentication Signal within 1 meter		
	Average Peak Power Level	Lower Frequency	Upper Frequency
1	-28 dBm	2.4149 GHz	2.430 GHz
2	-33 dBm	2.415 GHz	2.432 GHz
3	-35 dBm	2.4152 GHz	2.4315 GHz
4	-26 dBm	2.4147 GHz	2.4305 GHz
5	-30 dBm	2.4151 GHz	2.4318 GHz
6	-31 dBm	2.4148 GHz	2.4302 GHz
7	-25 dBm	2.4153 GHz	2.4317 GHz
8	-31 dBm	2.415 GHz	2.4306 GHz
9	-33 dBm	2.4149 GHz	2.4319 GHz
10	-29 dBm	2.4152 GHz	2.4308 GHz
11	-34 dBm	2.4148 GHz	2.4321 GHz
12	-37 dBm	2.4151 GHz	2.4316 GHz
13	-28 dBm	2.415 GHz	2.431 GHz
14	-31 dBm	2.4149 GHz	2.4315 GHz
15	-31 dBm	2.4153 GHz	2.4304 GHz
16	-32 dBm	2.415 GHz	2.4312 GHz
17	-30 dBm	2.4148 GHz	2.4309 GHz
18	-29 dBm	2.4151 GHz	2.4317 GHz
19	-26 dBm	2.4149 GHz	2.431 GHz
20	-30 dBm	2.4152 GHz	2.4314 GHz

21	-32 dBm	2.4147 GHz	2.4313 GHz
22	-30 dBm	2.415 GHz	2.4318 GHz
23	-28 dBm	2.4151 GHz	2.4306 GHz
24	-31 dBm	2.4148 GHz	2.4315 GHz
25	-30 dBm	2.4152 GHz	2.4317 GHz

Sample waveforms captured during the experiments are shown in the following figures. Figure 6 displays the waveform with the cursor measurement positioned at the center of the deauthentication signal.

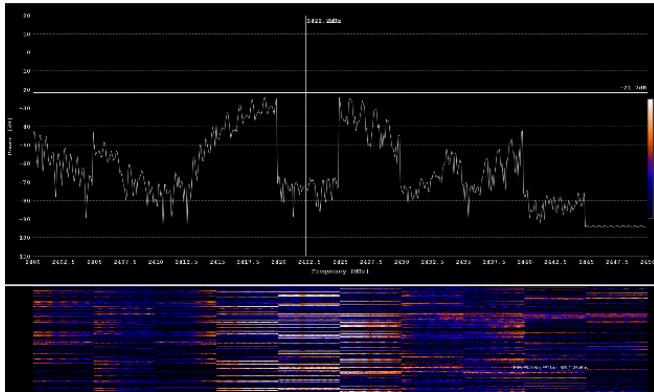


Fig. 6. Deauthentication signal captured within 1 meter (b)

B. Signal Strength Decrease with Distance

As the distance from the source of the attack increases, a corresponding decrease in signal strength is observed. At a distance of five meters, the signal strength reduces to -35 dBm, and at ten meters, it further decreases to -40 dBm. These measurements highlight the rapid attenuation of the attack signal with distance, providing crucial data for understanding the spatial impact of deauthentication attacks.

Table II presents the tabulated data captured when the Flipper Zero device is positioned within 5 meters of the SDR. The average peak power level observed is approximately -35 dBm, with the signal frequencies ranging from 2.41 GHz to 2.43 GHz

TABLE II. DEAUTHENTICATION SIGNAL DATA CAPTURED WITHIN 5 METERS

No	Deauthentication Signal within 5 meters		
	Average Peak Power Level	Lower Frequency	Upper Frequency
1	-35 dBm	2.4147 GHz	2.4303 GHz
2	-36 dBm	2.4151 GHz	2.4322 GHz
3	-34 dBm	2.4149 GHz	2.4317 GHz
4	-37 dBm	2.4146 GHz	2.4306 GHz
5	-35 dBm	2.4153 GHz	2.4315 GHz
6	-36 dBm	2.4145 GHz	2.4304 GHz
7	-33 dBm	2.4152 GHz	2.4318 GHz
8	-35 dBm	2.4148 GHz	2.4307 GHz

No	Deauthentication Signal within 5 meters		
	Average Peak Power Level	Lower Frequency	Upper Frequency
9	-36 dBm	2.4150 GHz	2.4314 GHz
10	-34 dBm	2.4147 GHz	2.4316 GHz
11	-37 dBm	2.4149 GHz	2.4321 GHz
12	-35 dBm	2.4151 GHz	2.4313 GHz
13	-36 dBm	2.4152 GHz	2.4308 GHz
14	-34 dBm	2.4146 GHz	2.4311 GHz
15	-37 dBm	2.4150 GHz	2.4317 GHz
16	-35 dBm	2.4148 GHz	2.4309 GHz
17	-36 dBm	2.4153 GHz	2.4312 GHz
18	-34 dBm	2.4147 GHz	2.4315 GHz
19	-37 dBm	2.4151 GHz	2.4306 GHz
20	-35 dBm	2.4149 GHz	2.4319 GHz
21	-36 dBm	2.4150 GHz	2.4307 GHz
22	-34 dBm	2.4146 GHz	2.4310 GHz
23	-37 dBm	2.4152 GHz	2.4318 GHz
24	-35 dBm	2.4149 GHz	2.4314 GHz
25	-36 dBm	2.4147 GHz	2.4305 GHz

Figure 7 shows the sample of captured deauthentication signals when flipper zero is placed 5 meters away from the SDR.

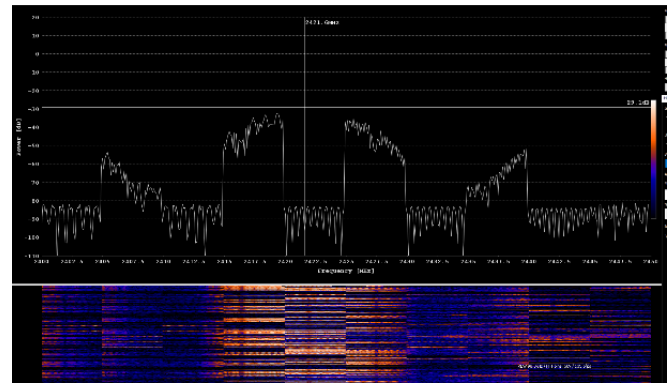


Fig. 7. Deauthentication signal captured within 5 meters

Table III presents the data captured when the Flipper Zero device is positioned within 10 meters of the SDR. The average peak power level observed is approximately -40 dBm, with the signal frequencies ranging from 2.41 GHz to 2.43 GHz.

TABLE III. DEAUTHENTICATION SIGNAL DATA CAPTURED WITHIN 10 METERS

No	Deauthentication Signal within 10 meters		
	Average Peak Power Level	Lower Frequency	Upper Frequency
1	-40 dBm	2.4145 GHz	2.4302 GHz
2	-41 dBm	2.4153 GHz	2.4325 GHz

3	-39 dBm	2.4147 GHz	2.4318 GHz
4	-42 dBm	2.4146 GHz	2.4307 GHz
5	-40 dBm	2.4152 GHz	2.4319 GHz
6	-41 dBm	2.4144 GHz	2.4306 GHz
7	-38 dBm	2.4151 GHz	2.4320 GHz
8	-40 dBm	2.4148 GHz	2.4310 GHz
9	-41 dBm	2.4149 GHz	2.4317 GHz
10	-39 dBm	2.4150 GHz	2.4316 GHz
11	-42 dBm	2.4147 GHz	2.4323 GHz
12	-40 dBm	2.4152 GHz	2.4314 GHz
13	-41 dBm	2.4153 GHz	2.4309 GHz
14	-39 dBm	2.4145 GHz	2.4311 GHz
15	-42 dBm	2.4151 GHz	2.4318 GHz
16	-40 dBm	2.4148 GHz	2.4313 GHz
17	-41 dBm	2.4150 GHz	2.4315 GHz
18	-39 dBm	2.4146 GHz	2.4312 GHz
19	-42 dBm	2.4153 GHz	2.4308 GHz
20	-40 dBm	2.4149 GHz	2.4317 GHz
21	-41 dBm	2.4151 GHz	2.4307 GHz
22	-39 dBm	2.4147 GHz	2.4310 GHz
23	-42 dBm	2.4152 GHz	2.4321 GHz
24	-40 dBm	2.4148 GHz	2.4315 GHz
25	-41 dBm	2.4145 GHz	2.4309 GHz

Figure 8 shows the sample of captured deauthentication signals when flipper zero is place 10 meters away from the SDR.

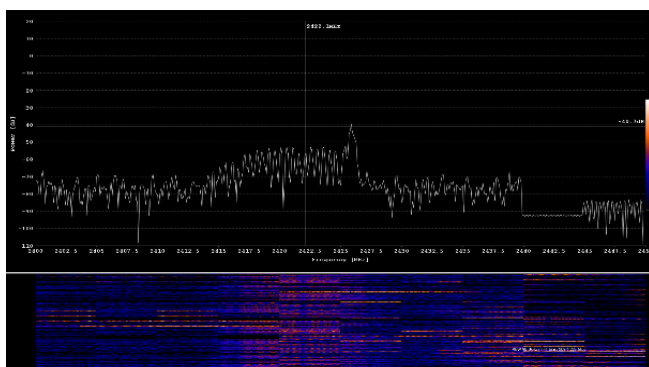


Fig. 8. Deauthentication signal data captured within 10 meters

The captured deauthentication signal, as shown in Figure 9, was recorded using a SDR set to I/Q modulation. The spectrum display indicates a significant signal spike at approximately 2.422 GHz. The waterfall display clearly shows a continuous red band at this frequency, indicating the persistent nature of the deauthentication packets.

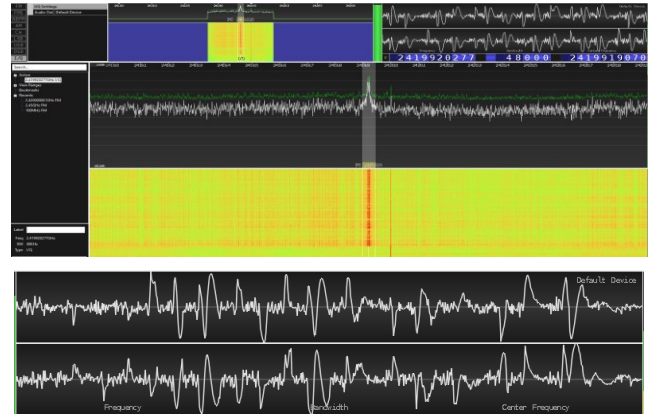


Fig. 9. Deauthentication signal captured using I/Q Modulation

The signal strength decreases by 5 dBm with increasing distance from 5 m to 10 m as show in figure 10 based on the data collected.

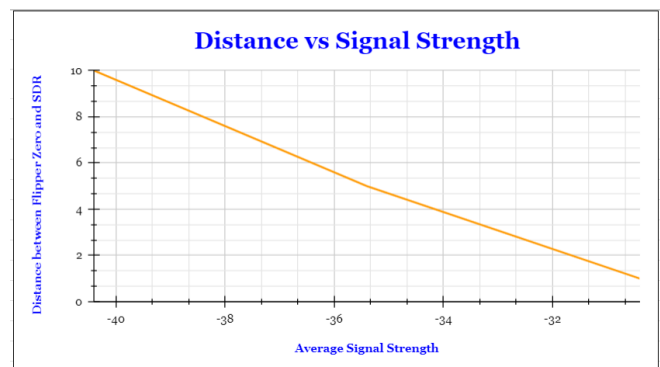


Fig. 10. Distance between Flipper Zero and SDR vs signal strength

This is supported by inverse square law, where it states that for a point source of waves that is capable of radiating omnidirectionally and with no obstructions in the vicinity, the intensity decreases with the square of the distance from the source [14].

$$I_{intensity} = \frac{1}{d^2_{distance}}$$

This means that when the distance from the source doubles, the intensity (power per unit area) reduces to 1/4 of the original intensity

In terms of dBm, a reduction in power to 1/4 of the original value corresponds to a 6 dB decrease

$$\Delta dBm = 10 \log_{10} \frac{P_{old}}{P_{new}}$$

$$\Delta dBm = 10 \log_{10} \frac{1}{4}$$

$$\Delta dBm = -6 \text{ dB theoretically}$$

From the tabulated data:

At 5 meters distance the average strength is -35.4 dBm

At 10 meters distance the average strength is -40.4 dBm

$$\Delta \text{ dBm} = -40.4 \text{ dBm} - (-35.4 \text{ dBm})$$

$$\Delta \text{ dBm} = -5 \text{ dB}$$

Based on the formula it predicts that the signal intensity should drop by a factor of 4 when the distance doubles. This corresponds to a 6 dB reduction in signal strength, which is almost the same as the calculated data.

CONCLUSION

This study investigates the characteristics of deauthentication attacks on wireless networks, with a specific focus on power level variations and signal properties. The problem addressed stems from the increasing availability of portable devices online, which can be easily acquired and used by novice hackers to launch deauthentication attacks. These attacks are often the first step in more advanced Wi-Fi attacks, as they disrupt the connection, enabling the capture of Wi-Fi packets or handshakes. This is particularly concerning for home networks, where WPA and WPA2 security protocols are still common, leaving them vulnerable to password cracking and further intrusion, while the adoption of WPA3 remains underutilized.

One significant issue is the bypassing of existing promiscuous mode detection tools when used alongside other hacking utilities. Therefore, new methods are needed to detect deauthentication attacks by analyzing specific parameters such as frequency range, interval, peak signal level, signal variation, and waveforms. These parameters are crucial to developing a device capable of detecting deauthentication attacks in real time and identifying the attack visually through spectral analysis.

Experimental results of this study shown considerable power level spikes from -80 dBm to -30 dBm at a distance of one meter, with a 6 MHz bandwidth centered at 2.422 GHz. As the distance increases, the signal strength drops to -35 dBm at five meters and -40 dBm at ten meters. These findings emphasize the importance of close-range monitoring and strategic sensor placement for effective detection of such attacks. The characteristic bandwidth of the deauthentication signal provides a unique spectral signature, offering a reliable means of identification. By integrating waveform analysis and advanced signal processing techniques with existing packet-based methods, more robust and effective detection systems can be developed, thereby improving the resilience of wireless networks against deauthentication attacks.

As a step forward, the author is currently developing a WLAN protection device aimed at safeguarding home networks from various types of attacks, including deauthentication attacks. One of the key requirements for this device is the ability to detect and alert users to the presence of potential deauthentication attacks nearby, offering a critical layer of defense. The next phase of research will focus on refining detection parameters such as signal strength, frequency range, and waveform characteristics. Additionally, the device will incorporate a mobile

application to notify users in real time, providing actionable insights and protection against wireless network threats. Further exploration of detection algorithms and the integration of advanced signal processing techniques will be essential to the success of this system, paving the way for more secure home networks in the future.

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